THE GAUSSIAN PRODUCT INEQUALITY FOR DISJOINT PRINCIPAL MINORS OF WISHART MATRICES

KNOWLEDGE SENSE

Abstract. Let W be a real $p \times p$ Wishart matrix with real shape parameter $\alpha > p-1$ and positive definite scale matrix, and let $W_{11}, \dots, W_{dd}$ be its diagonal blocks for a partition of p into positive integers. We prove that $\mathsf{E}[\prod_i \det(W_{ii})^{\nu_i}] \geq \prod_i \mathsf{E}[\det(W_{ii})^{\nu_i}]$ for all real $\nu_1, \dots, \nu_d \geq 0$, which settles a conjecture of Genest, Ouimet and Richards; the case $d = 2$ was known. The proof separates one diagonal block at a time. Conditionally on W_{11}, the remaining blocks form a noncentral Wishart matrix whose noncentrality matrix is a fixed congruence transform of W_{11}. We show that the expected product of determinant powers is nondecreasing in the noncentrality matrix for the Loewner order, provided that its range stays in a fixed subspace. This rests on a second-order differential equation in a rank-one noncentrality parameter, valid for every real shape parameter larger than the order of the matrix, and on the pointwise positivity of an associated second-order differential operator on products of determinant powers. The separation is completed by the association of the eigenvalues of a Wishart matrix with identity scale matrix.

1. Introduction

Let W be a real $p \times p$ Wishart matrix with α degrees of freedom and positive definite scale matrix Σ, written $W \sim \mathcal{W}_p(\alpha, \Sigma)$; here $\alpha > p-1$ is real. Partition $W = (W_{ij})_{1 \leq i,j \leq d}$ into blocks of sizes $p_i \times p_j$, where $p_1 + \cdots + p_d = p$, and write $|W_{ii}|$ for the determinant of the diagonal block W_{ii}. In 1934 Wilks [14] asked for the joint moments $\mathsf{E}[\prod_i |W_{ii}|^{\nu_i}]$ of these disjoint principal minors and described the general case as "extremely complicated". For $d = 2$, Genest, Ouimet and Richards [9] expressed this moment through a Gaussian hypergeometric function of matrix argument. For nonnegative integer exponents and real α, Gaunt and Ouimet [6] gave a finite exact recursion based on a Stein identity for the Wishart distribution [1]. For embedded (nested) principal minors, explicit formulas go back to Wilks; see [8].

The Gaussian product inequality asserts that $\mathsf{E}[\prod_i |X_i|^{\gamma_i}] \geq \prod_i \mathsf{E}[|X_i|^{\gamma_i}]$ for every centered Gaussian vector $(X_1, \dots, X_n)$ and all $\gamma_i > 0$. It grew out of the real linear polarization constant problem of Benítez, Sarantopoulos and Tonge [2]; Frenkel [5] formulated it for even integer exponents, Li and Wei [11] proposed the version with real exponents, and a proof has recently been posted by Ouimet and Greaves [13]. Genest, Ouimet and Richards [7, Conjecture 1.1] proposed the following matrix analogue, in which squared Gaussian coordinates are replaced by disjoint principal minors of a Wishart matrix.

Date: 8 October 2026.

2020 *Mathematics Subject Classification.* Primary 60E15; Secondary 62H10, 15A15, 60E05.

Key words and phrases. Gaussian product inequality, Wishart distribution, principal minors, noncentral Wishart distribution, Loewner order, total positivity.

Theorem 1.1. *Let $d, p_1, \dots, p_d$ be positive integers, $p = p_1 + \cdots + p_d$, let $\alpha > p-1$ be real and let Σ be a real positive definite $p \times p$ matrix. Let $W \sim \mathcal{W}_p(\alpha, \Sigma)$ be partitioned into blocks W_{ij} of sizes $p_i \times p_j$. Then*

$$\mathsf{E}\Big[\prod_{i=1}^{d} |W_{ii}|^{\nu_i}\Big] \geq \prod_{i=1}^{d} \mathsf{E}\big[|W_{ii}|^{\nu_i}\big] \qquad \text{for all real } \nu_1, \dots, \nu_d \geq 0. \tag{1.1}$$

For $d = 2$, Theorem 1.1 is [9, Corollary 3.4], which is in fact a quantitative strengthening. Genest, Ouimet and Richards [7] also proved product inequalities for nonpositive exponents and for some mixtures of signs, one of them conditionally on the conjecture, and a product inequality for nondecreasing functions of the eigenvalues of W [7, Theorem 3.14]. When all blocks are scalar and α is an integer, (1.1) can be derived from the strong Gaussian product inequality [13, Theorem 1]; see Remark 7.4.

The conjecture concerns all real $\alpha > p-1$, the full range in which $\mathcal{W}_p(\alpha, \Sigma)$ has a density, and all real exponents $\nu_1, \dots, \nu_d \geq 0$. Real exponents are what one needs to differentiate (1.1) in the ν_i or to combine it with inverse powers as in [7, Proposition 3.9]. This generality is also what makes the case $d \geq 3$ difficult. The hypergeometric representation of [9] has no known analogue for three or more blocks; the recursion of [6] is exact for integer exponents but does not terminate for real ones [6, Remark 3.2]; and the iterated Schur complements that handle embedded minors [8] do not apply, because removing one diagonal block by a Schur complement changes the remaining disjoint blocks instead of producing independent factors. We are not aware of a previous proof for $d \geq 3$. We deduce Theorem 1.1 from the following statement.

Theorem 1.2. *In the setting of Theorem 1.1, inequality* (1.1) *holds in each of the following cases:*

(a) $\alpha \geq p$;
(b) $d \geq 3$;
(c) $d = 2$, *and either* $p_1 \neq p_2$ *or* $\operatorname{rank} \Sigma_{12} < p_1$.

Moreover, in cases (a) and (b),

$$\mathsf{E}\Big[\prod_{i=1}^{d} |W_{ii}|^{\nu_i}\Big] \geq \mathsf{E}\big[|W_{11}|^{\nu_1}\big]\, \mathsf{E}\Big[\prod_{i=2}^{d} |W_{ii}|^{\nu_i}\Big]. \tag{1.2}$$

Theorem 1.2 leaves out only the case $d = 2$, $p_1 = p_2$, Σ_{12} invertible and $p - 1 < \alpha < p$, which is covered by [9, Corollary 3.4]. The proof works directly with real α and real exponents; it uses neither an explicit formula for the joint moments nor interpolation from integer parameters.

Outline of the proof. After a block diagonal congruence we may assume that $\Sigma_{11} = I$. Conditionally on W_{11}, the principal submatrix formed by the blocks $2, \dots, d$ has a noncentral Wishart law with shape α, a fixed scale matrix and noncentrality matrix $K^{\mathsf{T}} W_{11} K$, where $K = (\Sigma_{12}, \dots, \Sigma_{1d})$ (Lemma 6.1). Let $G(W_{11})$ be the conditional expectation of $\prod_{i \geq 2} |W_{ii}|^{\nu_i}$. If G is nondecreasing in the Loewner order, then its average over orthogonal conjugations is a coordinatewise nondecreasing function of the eigenvalues of W_{11}. These eigenvalues have a multivariate totally positive density and are therefore associated, which gives (1.2) (Proposition 6.3).

Iterating (1.2), and choosing the separated block suitably when $d = 2$, proves Theorem 1.2.

The main step is the monotonicity of G. All the noncentrality matrices $K^\mathsf{T} W_{11} K$ have their range in the subspace $L = \operatorname{ran} K^\mathsf{T}$, of dimension $r = \operatorname{rank} K$. Let Ω have its range in L, and let $v \in L$. We write $\Omega = \Omega' + \theta_0 vv^\mathsf{T}$ with $\operatorname{rank} \Omega' \le r - 1$, so that $\Omega + c\, vv^\mathsf{T} = \Omega' + (\theta_0 + c)vv^\mathsf{T}$. The part Ω' is carried by $r - 1$ Gaussian rows, and this leaves the shape $q = \alpha - r + 1$ for a noncentral Wishart matrix W_θ with noncentrality matrix θvv^T. Let f be a product of nonnegative powers of the determinants of the diagonal blocks of W_θ, each shifted by the corresponding block of the Gaussian part. Then $g(\theta) = \mathsf{E} f(W_\theta)$ satisfies

$$2\theta g''(\theta) + q g'(\theta) = \mathsf{E}\big[\mathcal{R}_{q,vv^\mathsf{T}} f(W_\theta)\big]$$

(Lemma 4.2), where $\mathcal{R}_{q,H}$ is a second-order differential operator that reduces for $(q, H) = (\alpha, \Sigma)$ to the Wishart Stein operator of [1, 6]. By Lemma 3.1, $\mathcal{R}_{q,vv^\mathsf{T}} f \ge 0$ on the positive semidefinite cone whenever $q \ge \ell + 1$, where ℓ is the largest block size among the blocks $2, \dots, d$. Hence g is nondecreasing (Proposition 4.4), and G is monotone (Proposition 5.1). The resulting condition $q \ge \ell + 1$ reads $\alpha \ge r + \ell$. It holds whenever $d \ge 3$, and for $d = 2$ after a suitable choice of the block to be separated, except in the case left out of Theorem 1.2.

Besides standard facts on Wishart laws [12], the proof uses Fourier inversion on the space of symmetric matrices and the association of random vectors with a multivariate totally positive density [4, 10].

Organization. Section 2 collects facts on Laplace transforms and noncentral Wishart laws. Section 3 computes the operator $\mathcal{R}_{q,H}$ on products of determinant powers. Section 4 proves the differential equation in a rank-one noncentrality parameter and the resulting monotonicity, and Section 5 extends it to noncentrality matrices with a fixed range. Section 6 separates one block, and Section 7 proves the theorems and adds some remarks.

2. Preliminaries

2.1. **Notation.** For $m \in \mathbb{N}$, let $\mathcal{S}^m$ be the space of real symmetric $m \times m$ matrices, with inner product $\langle U, W \rangle = \operatorname{tr}(UW)$ and Frobenius norm $\|\cdot\|$, and let $\mathcal{S}^m_+$ and $\mathcal{S}^m_{++}$ be the cones of positive semidefinite and positive definite matrices. We write $A \preceq B$ if $B - A \in \mathcal{S}^m_+$, $|A|$ for the determinant, and $\operatorname{etr}(A) = \exp(\operatorname{tr} A)$. On $\mathcal{S}^m_+$ we have $\|W\| \le \operatorname{tr} W \le \sqrt{m}\, \|W\|$, and for $B \in \mathcal{S}^r_+$ the arithmetic and geometric mean inequality gives $|B| \le (\operatorname{tr} B / r)^r$. Lebesgue measure on $\mathcal{S}^m$ refers to the coordinates $(w_{ab})_{a \le b}$.

For a function f of class C^2 on an open subset of $\mathcal{S}^m$ we use the symmetric gradient $\nabla f = (\nabla_{ab} f)$, where $\nabla_{ab} = \nabla_{ba} = \frac{1}{2}(1 + \delta_{ab})\, \partial / \partial w_{ab}$ for $a \le b$. Thus $\langle \nabla f(W), U \rangle$ is the derivative of f at W in the direction U. For $q \in \mathbb{R}$ and $H \in \mathcal{S}^m$ we define

$$\mathcal{R}_{q,H} f(W) = q \operatorname{tr}\big(H \nabla f(W)\big) + 2 \operatorname{tr}\big(W \nabla H \nabla f(W)\big), \tag{2.1}$$

where

$$\operatorname{tr}(W \nabla H \nabla f) = \sum_{i,j,k,l=1}^{m} W_{ik} H_{lj} \nabla_{ij} \nabla_{kl} f.$$

For $T \in \mathcal{S}^m$ one checks that $\nabla \operatorname{etr}(-TW) = -T \operatorname{etr}(-TW)$, and hence

$$\mathcal{R}_{q,H} \operatorname{etr}(-TW) = \big(-q \operatorname{tr}(HT) + 2 \operatorname{tr}(TWTH)\big) \operatorname{etr}(-TW). \tag{2.2}$$

The same formula holds for complex symmetric T. For $(q, H) = (\alpha, \Sigma)$, (2.1) is the operator $\mathcal{R}_{\alpha,\Sigma}$ of [6, (3.2)], and the Stein identity of [1], in the form [6, (3.1)], states that $\mathsf{E}[\mathcal{R}_{\alpha,\Sigma} f(W)] = \mathsf{E}[\operatorname{tr}(W \nabla f(W))]$ for $W \sim \mathcal{W}_m(\alpha, \Sigma)$ and polynomial f.

Blocks of a matrix are indexed as in Theorem 1.1. We write ran for the range of a matrix and Ω^+ for the Moore–Penrose inverse of $\Omega \in \mathcal{S}_+^m$.

2.2. **Laplace transforms.** The following lemma lets us pass from exponential test functions to compactly supported ones. With $\mathcal{R} = 0$ it says that a finite signed measure on $\mathcal{S}_+^m$ is determined by its Laplace transform on $\mathcal{S}_{++}^m$.

Lemma 2.1. *Let μ be a finite positive Borel measure on $\mathcal{S}_+^m$ with $\int (1 + \operatorname{tr} W)\, \mu(dW) < \infty$, let ξ be a finite signed Borel measure on $\mathcal{S}_+^m$, and let $\mathcal{R}$ be either 0 or $\mathcal{R}_{q,H}$ for some $q \in \mathbb{R}$ and $H \in \mathcal{S}^m$. Suppose that*

$$\int \operatorname{etr}(-TW)\, \xi(dW) = \int \mathcal{R}\big[\operatorname{etr}(-T\,\cdot)\big](W)\, \mu(dW) \qquad \text{for all } T \in \mathcal{S}_{++}^m. \tag{2.3}$$

Then $\int \varphi \, d\xi = \int \mathcal{R}\varphi \, d\mu$ for every $\varphi \in C_c^\infty(\mathcal{S}^m)$.

Proof. Let $\mathcal{T}$ be the set of complex symmetric matrices T with $\operatorname{Re} T \in \mathcal{S}_{++}^m$. For $T \in \mathcal{T}$ and $W \in \mathcal{S}_+^m$ we have $|\operatorname{etr}(-TW)| = \operatorname{etr}(-(\operatorname{Re} T)W) \le 1$, and by (2.2) the integrand on the right of (2.3) is bounded by $c(1 + \|T\|)^2(1 + \|W\|)$. Hence both sides of (2.3) are holomorphic functions on $\mathcal{T}$, since the integrands are entire in the entries of T and locally uniformly dominated by integrable functions. Their difference vanishes on the open set $\mathcal{S}_{++}^m$ of real points of $\mathcal{T}$, so all its complex partial derivatives vanish at a point of $\mathcal{S}_{++}^m$; as $\mathcal{T}$ is convex, hence connected, the difference vanishes on $\mathcal{T}$.

Fix $T_0 \in \mathcal{S}_{++}^m$ and let $\varphi \in C_c^\infty(\mathcal{S}^m)$. Then $\psi(W) = \operatorname{etr}(T_0 W)\varphi(W)$ belongs to $C_c^\infty(\mathcal{S}^m)$, and its Fourier transform $\hat{\psi}(U) = \int \psi(W) e^{-i\langle U, W\rangle} dW$ is a Schwartz function. Fourier inversion gives a constant $c_m > 0$ with

$$\varphi(W) = c_m \int_{\mathcal{S}^m} \hat{\psi}(U) \operatorname{etr}\big(-(T_0 - iU)W\big)\, dU, \qquad W \in \mathcal{S}^m.$$

Differentiation under the integral sign shows that

$$\mathcal{R}\varphi(W) = c_m \int_{\mathcal{S}^m} \hat{\psi}(U)\, \mathcal{R}\big[\operatorname{etr}(-(T_0 - iU)\,\cdot)\big](W)\, dU.$$

For $W \in \mathcal{S}_+^m$ the last integrand is bounded by $c|\hat{\psi}(U)|(1+\|U\|)^2(1+\|W\|)$, which is integrable with respect to $dU\, \mu(dW)$; likewise $|\hat{\psi}(U) \operatorname{etr}(-(T_0 - iU)W)| \le |\hat{\psi}(U)|$. Since $T_0 - iU \in \mathcal{T}$, Fubini's theorem and the identity on $\mathcal{T}$ give $\int \varphi \, d\xi = \int \mathcal{R}\varphi \, d\mu$. □

2.3. **Wishart and noncentral Wishart laws.** For $\alpha > m - 1$ and $\Sigma \in \mathcal{S}_{++}^m$, the Wishart law $\mathcal{W}_m(\alpha, \Sigma)$ has the density

$$\frac{|W|^{(\alpha - m - 1)/2} \operatorname{etr}(-\Sigma^{-1} W/2)}{2^{\alpha m/2} |\Sigma|^{\alpha/2} \Gamma_m(\alpha/2)}, \qquad W \in \mathcal{S}_{++}^m,$$

where $\Gamma_m(a) = \pi^{m(m-1)/4} \prod_{j=1}^m \Gamma(a - (j-1)/2)$. Its Laplace transform is $\mathsf{E} \operatorname{etr}(-TW) = |I + 2\Sigma T|^{-\alpha/2}$ for $T \in \mathcal{S}_+^m$, more generally $\mathsf{E} \operatorname{etr}(TW) =$

$|I - 2T\Sigma|^{-\alpha/2}$ whenever $\Sigma^{-1} - 2T \in \mathcal{S}_{++}^m$ [12, Theorem 3.2.3], and its mean is $\alpha\Sigma$. In particular $\mathsf{E}\exp(t \operatorname{tr} W) < \infty$ for some $t > 0$, so $\operatorname{tr} W$ has moments of all orders. Since $|W_{ii}| \le (\operatorname{tr} W)^{p_i}$ on $\mathcal{S}_+^p$, every product of nonnegative powers of the determinants of diagonal blocks of a Wishart matrix has finite expectation. By Lemma 2.1 and the Laplace transform, if $W \sim \mathcal{W}_m(\alpha, \Sigma)$ and M is an invertible $m \times m$ matrix, then $MWM^{\mathsf{T}} \sim \mathcal{W}_m(\alpha, M\Sigma M^{\mathsf{T}})$, and a principal submatrix W_{II} satisfies $W_{II} \sim \mathcal{W}_{|I|}(\alpha, \Sigma_{II})$.

Fix $S \in \mathcal{S}_{++}^m$ and $T \in \mathcal{S}_+^m$, and put

$$B_T = (I + 2ST)^{-1}, \qquad S_T = (S^{-1} + 2T)^{-1} = B_T S. \tag{2.4}$$

Then $TB_T = B_T^{\mathsf{T}} T$ is symmetric, because $(I + 2TS)T = T(I + 2ST)$, and

$$T - 2TS_T T = T - 2TB_T ST = TB_T\big((I + 2ST) - 2ST\big) = TB_T. \tag{2.5}$$

Definition 2.2. Let $\beta \ge 0$, $S \in \mathcal{S}_{++}^m$ and $\Omega \in \mathcal{S}_+^m$. A random matrix Y with values in $\mathcal{S}_+^m$ is of *type* (β, S, Ω) if

$$\mathsf{E}\operatorname{etr}(-TY) = |I + 2ST|^{-\beta/2} \operatorname{etr}(-\Omega T B_T) \qquad \text{for all } T \in \mathcal{S}_+^m. \tag{2.6}$$

If $\beta > m - 1$ and a law of type (β, S, Ω) exists, we denote it by $\mathcal{NW}_m(\beta, S, \Omega)$. We call Ω the noncentrality matrix. In the representation of Lemma 2.3 below, $\Omega = M^{\mathsf{T}} M$ is the Gram matrix of the mean matrix M; it is not multiplied by S^{-1}.

By Lemma 2.1, the type determines the law. If Y_1 and Y_2 are independent of types (β_1, S, Ω_1) and (β_2, S, Ω_2), then $Y_1 + Y_2$ is of type $(\beta_1 + \beta_2, S, \Omega_1 + \Omega_2)$. A matrix with law $\mathcal{W}_m(\beta, S)$ is of type $(\beta, S, 0)$.

Lemma 2.3. *Let $S \in \mathcal{S}_{++}^m$, $T \in \mathcal{S}_+^m$ and $\mu \in \mathbb{R}^m$. Then, as measures in $x \in \mathbb{R}^m$,*

$$e^{-x^{\mathsf{T}} T x}\, N_m(\mu, S)(dx) = |I + 2ST|^{-1/2} e^{-\mu^{\mathsf{T}} T B_T \mu}\, N_m(B_T\mu, S_T)(dx). \tag{2.7}$$

Consequently, if $n \ge 0$ is an integer, M is a real $n \times m$ matrix and Z is an $n \times m$ matrix with independent $N_m(0, S)$ rows, then $(Z+M)^{\mathsf{T}}(Z+M)$ is of type $(n, S, M^{\mathsf{T}}M)$; for $n = 0$ this matrix is 0. In particular $\mathcal{NW}_m(\beta, S, \Omega)$ exists whenever $\operatorname{rank}\Omega \le n$ and $\beta - n > m - 1$, and it is the law of $(Z + M)^{\mathsf{T}}(Z + M) + V$ for any M with $M^{\mathsf{T}}M = \Omega$ and $V \sim \mathcal{W}_m(\beta - n, S)$ independent of Z.

Proof. Using $S_T^{-1} = S^{-1} + 2T$, $S_T^{-1} B_T = S^{-1}$ and $B_T^{\mathsf{T}} S^{-1} + 2TB_T = S^{-1}B_T + 2TB_T = S^{-1}$, one obtains

$$x^{\mathsf{T}} T x + \tfrac{1}{2}(x - \mu)^{\mathsf{T}} S^{-1}(x - \mu) = \tfrac{1}{2}(x - B_T\mu)^{\mathsf{T}} S_T^{-1}(x - B_T\mu) + \mu^{\mathsf{T}} T B_T \mu.$$

Since $|S_T|/|S| = |I + 2ST|^{-1}$, this proves (2.7). For the Gram matrix, $\operatorname{etr}(-T(Z + M)^{\mathsf{T}}(Z+M))$ is the product over the rows x_i^{T} of $Z+M$ of $e^{-x_i^{\mathsf{T}} T x_i}$, and (2.7) applied to each row gives the type $(n, S, M^{\mathsf{T}}M)$. The last assertion follows by adding an independent matrix of type $(\beta - n, S, 0)$. □

3. A positive second-order operator

Throughout this section, $m = m_1 + \cdots + m_s$ is a partition of m into positive integers, $W \in \mathcal{S}^m$ is partitioned accordingly into blocks W_{jj} of size m_j, and $\ell = \max_j m_j$. Let $K_j \in \mathcal{S}_{++}^{m_j}$ and $b_j \ge 0$, and put

$$\mathcal{O} = \{W \in \mathcal{S}^m : W_{jj} + K_j \in \mathcal{S}_{++}^{m_j} \text{ for all } j\}, \qquad f(W) = \prod_{j=1}^{s} |W_{jj} + K_j|^{b_j}.$$

The set $\mathcal{O}$ is open and contains $\mathcal{S}_+^m$, and f is smooth and positive on $\mathcal{O}$. For $W \in \mathcal{O}$ let $A_j = W_{jj} + K_j$, and put

$$D = \operatorname{diag}_j\big(b_j A_j^{-1}\big), \qquad h_j = \operatorname{tr}\big(A_j^{-1} W_{jj}\big), \qquad Q_j = A_j^{-1} W_{jj} A_j^{-1}.$$

For $H \in \mathcal{S}^m$ and $v \in \mathbb{R}^m$, we write H_{jj} and v_j for the corresponding diagonal block and subvector.

Lemma 3.1. *For every $q \in \mathbb{R}$, $H \in \mathcal{S}^m$ and $W \in \mathcal{O}$,*

$$\frac{\mathcal{R}_{q,H} f(W)}{f(W)} = 2\operatorname{tr}(HDWD) + \sum_{j=1}^{s} b_j \Big((q - h_j) \operatorname{tr}\big(H_{jj} A_j^{-1}\big) - \operatorname{tr}\big(H_{jj} Q_j\big) \Big). \tag{3.1}$$

In particular, for $v \in \mathbb{R}^m$,

$$\frac{\mathcal{R}_{q,vv^{\mathsf{T}}} f(W)}{f(W)} = 2v^{\mathsf{T}} DWDv + \sum_{j=1}^{s} b_j \, v_j^{\mathsf{T}} \big((q - h_j) A_j^{-1} - Q_j \big) v_j. \tag{3.2}$$

If $W \in \mathcal{S}_+^m$ and $q \geq \ell + 1$, then $\mathcal{R}_{q,vv^{\mathsf{T}}} f(W) \geq 0$ for every $v \in \mathbb{R}^m$.

Proof. For a, b in the jth block, $\nabla_{ab} \log |A_j| = (A_j^{-1})_{ab}$, both for $a = b$ and for $a \neq b$; hence $\nabla f = fD$. Differentiating A_j^{-1} along the symmetric coordinate directions gives, for a, b, c, d in the jth block,

$$\nabla_{ab} (A_j^{-1})_{cd} = -\tfrac{1}{2} \big((A_j^{-1})_{ca} (A_j^{-1})_{bd} + (A_j^{-1})_{cb} (A_j^{-1})_{ad} \big),$$

again including the case $a = b$, while $\nabla_{ab} D_{cd} = 0$ unless a, b, c, d lie in one block. Therefore

$$\frac{\nabla_{ij} \nabla_{kl} f}{f} = D_{ij} D_{kl} - \sum_{t=1}^{s} \frac{b_t}{2} \mathbf{1}_{\{i,j,k,l \in \mathcal{I}_t\}} \big((A_t^{-1})_{ki} (A_t^{-1})_{jl} + (A_t^{-1})_{kj} (A_t^{-1})_{il} \big),$$

where $\mathcal{I}_t$ denotes the set of indices of the tth block. Contracting with $W_{ik} H_{lj}$, the first term gives $\operatorname{tr}(DWDH)$, and the two terms in parentheses give $\operatorname{tr}(A_t^{-1} W_{tt}) \operatorname{tr}(H_{tt} A_t^{-1})$ and $\operatorname{tr}(A_t^{-1} W_{tt} A_t^{-1} H_{tt})$. Together with $q \operatorname{tr}(H \nabla f)/f = q \sum_j b_j \operatorname{tr}(H_{jj} A_j^{-1})$ this proves (3.1), and (3.2) is the case $H = vv^{\mathsf{T}}$.

Let $W \in \mathcal{S}_+^m$. Then $0 \preceq W_{jj} \preceq A_j$, so $X_j = A_j^{-1/2} W_{jj} A_j^{-1/2}$ satisfies $0 \preceq X_j \preceq I$. Hence $0 \leq h_j = \operatorname{tr} X_j \leq m_j$ and $0 \preceq Q_j = A_j^{-1/2} X_j A_j^{-1/2} \preceq A_j^{-1}$. If $q \geq \ell + 1$, then

$$(q - h_j) A_j^{-1} - Q_j \succeq (q - m_j - 1) A_j^{-1} \succeq 0,$$

and $DWD \succeq 0$. Since $f > 0$ and $b_j \geq 0$, the right side of (3.2) is nonnegative. □

Lemma 3.2. *Let $\kappa = \min_j \lambda_{\min}(K_j)$ and $\beta = \sum_j m_j b_j$. There is $\tilde{f} \in C^\infty(\mathcal{S}^m)$ that coincides with f on a neighborhood of $\mathcal{S}_+^m$ and satisfies $|\partial^\gamma \tilde{f}(W)| \leq c(1 + \|W\|)^\beta$ for all $W \in \mathcal{S}^m$ and all partial derivatives ∂^γ of order at most two. Moreover, $|\mathcal{R}_{q,H} f(W)| \leq c(1 + \|W\|)^{\beta+1}$ for $W \in \mathcal{S}_+^m$.*

Proof. Let $\delta = \kappa/6$ and $N_t = \{W : \operatorname{dist}(W, \mathcal{S}_+^m) < t\}$. If $\operatorname{dist}(W, \mathcal{S}_+^m) \leq 3\delta$, there is $W' \in \mathcal{S}_+^m$ with $\|W - W'\| \leq 3\delta$, so $A_j \succeq W'_{jj} + K_j - 3\delta I \succeq (\kappa/2) I$; thus the closure of $N_{3\delta}$ lies in $\mathcal{O}$, and on it $\|A_j^{-1}\| \leq 2/\kappa$ and $|A_j| \leq (\operatorname{tr} A_j / m_j)^{m_j} \leq c(1 + \|W\|)^{m_j}$. By $\nabla f = fD$ and the formula for $\nabla\nabla f$ above, f and its derivatives of order at most two are bounded by $c(1 + \|W\|)^\beta$ there. Let ζ be the convolution of the indicator function of $N_{2\delta}$ with a smooth probability density supported in the ball of radius δ. Then ζ is smooth with bounded derivatives, $\zeta = 1$ on N_δ, and $\zeta = 0$ outside $N_{3\delta}$.

Put $\tilde f = \zeta f$ on $\mathcal{O}$ and $\tilde f = 0$ elsewhere. The bound on $\mathcal{R}_{q,H} f$ follows from (3.1), because $0 \le h_j \le m_j$, $\|Q_j\| \le \|A_j^{-1}\| \le 1/\kappa$ and $\|DWD\| \le c\|W\|$ on $\mathcal{S}_+^m$. □

4. A rank-one noncentrality parameter

Throughout this section, $S \in \mathcal{S}_{++}^m$, $v \in \mathbb{R}^m$ and $q > m$ is real. Let $X_0 \sim N_m(0, S)$ and $V \sim \mathcal{W}_m(q-1, S)$ be independent; V exists since $q-1 > m-1$. For $s \in \mathbb{R}$ put

$$W^{(s)} = (X_0 + sv)(X_0 + sv)^{\mathsf T} + V,$$

and for $\theta \ge 0$ let μ_θ be the law of $W^{(\sqrt\theta)}$. By Lemma 2.3, $\mu_\theta = \mathcal{NW}_m(q, S, \theta v v^{\mathsf T})$ and

$$L_\theta(T) := \int \operatorname{etr}(-TW)\, \mu_\theta(dW) = |I + 2ST|^{-q/2} e^{-\theta a_T}, \qquad a_T = v^{\mathsf T} T B_T v. \tag{4.1}$$

For an integer $N \ge 0$, let $\mathcal{M}_N$ be the Banach space of finite signed Borel measures η on $\mathcal{S}_+^m$ with $\|\eta\|_N = \int (1 + \operatorname{tr} W)^N\, |\eta|(dW) < \infty$.

Lemma 4.1. *There are measures $\mu_\theta \in \bigcap_N \mathcal{M}_N$, defined for all real θ and agreeing with the laws above for $\theta \ge 0$, such that $\theta \mapsto \mu_\theta$ is given on $\mathbb{R}$ by a power series that converges in $\mathcal{M}_N$ for every N and every θ. Consequently, if f is a Borel function on $\mathcal{S}_+^m$ with $|f(W)| \le c(1 + \operatorname{tr} W)^N$, then $g(\theta) = \int f\, d\mu_\theta$ is real analytic on $\mathbb{R}$ and $g^{(k)}(\theta) = \int f\, d\mu_\theta^{(k)}$ for all k, where $\mu_\theta^{(k)}$ is the termwise derivative.*

Proof. Let $a = v^{\mathsf T} S^{-1} v$ and $u(x) = v^{\mathsf T} S^{-1} x$, and define polynomials P_n by $e^{su - as^2/2} = \sum_{n \ge 0} P_n(u) s^n / n!$. Since the left side is invariant under $(s, u) \mapsto (-s, -u)$, we have $P_n(-u) = (-1)^n P_n(u)$. Cauchy's estimate on the circle $|s| = \rho$ gives $|P_n(u)| \le n!\, \rho^{-n} e^{\rho|u| + a\rho^2/2}$. Put $F(x, Y) = xx^{\mathsf T} + Y$ and define $\eta_n(\varphi) = \mathsf{E}[\varphi(F(X_0, V)) P_n(u(X_0))]$. As $x \mapsto -x$ preserves the law of X_0 and fixes F, we get $\eta_n = 0$ for odd n. Moreover $(1 + \operatorname{tr} F(x, Y))^N \le (1 + \|x\|^2)^N (1 + \operatorname{tr} Y)^N$, so

$$\|\eta_n\|_N \le n!\, \rho^{-n} M_{N,\rho},$$

where

$$M_{N,\rho} = e^{a\rho^2/2}\, \mathsf{E}\big[(1 + \|X_0\|^2)^N e^{\rho |u(X_0)|}\big]\, \mathsf{E}\big[(1 + \operatorname{tr} V)^N\big] < \infty.$$

Hence $\mu_\theta := \sum_{k \ge 0} \theta^k \eta_{2k} / (2k)!$ converges in $\mathcal{M}_N$ for $|\theta| < \rho^2$, and ρ is arbitrary. The density of $N_m(sv, S)$ with respect to $N_m(0, S)$ is $e^{su(x) - as^2/2}$, so for $\theta = s^2 \ge 0$ and bounded Borel φ, dominated convergence gives

$$\mathsf{E}\varphi(W^{(s)}) = \mathsf{E}\big[\varphi(F(X_0, V)) e^{su(X_0) - as^2/2}\big] = \sum_{n \ge 0} \frac{s^n}{n!} \eta_n(\varphi) = \int \varphi\, d\mu_\theta.$$

The last statement follows because $\eta \mapsto \int f\, d\eta$ is a bounded linear functional on $\mathcal{M}_N$. □

For $\theta \ge 0$ put $\xi_\theta = 2\theta \mu_\theta'' + q \mu_\theta'$, a finite signed measure in $\bigcap_N \mathcal{M}_N$.

Lemma 4.2. *For every $\theta \ge 0$ and $\varphi \in C_c^\infty(\mathcal{S}^m)$,*

$$\int \varphi\, d\xi_\theta = \int \mathcal{R}_{q, vv^{\mathsf T}} \varphi\, d\mu_\theta. \tag{4.2}$$

Proof. By Lemma 2.1, applied with $\xi = \xi_\theta$, $\mu = \mu_\theta$ and $\mathcal{R} = \mathcal{R}_{q,vv^\intercal}$, it suffices to verify (2.3), that is,

$$\int \operatorname{etr}(-TW)\,\xi_\theta(dW) = \int \mathcal{R}_{q,vv^\intercal}\big[\operatorname{etr}(-T\,\cdot\,)\big](W)\,\mu_\theta(dW) \qquad \text{for all } T \in \mathcal{S}^m_{++}.$$

Fix $T \in \mathcal{S}^m_{++}$. By Lemma 4.1, $\theta \mapsto \int \operatorname{etr}(-TW)\,\mu_\theta(dW)$ is real analytic on $\mathbb{R}$; it agrees with $|I+2ST|^{-q/2}e^{-\theta a_T}$ for $\theta \ge 0$, hence for all real θ. By Lemma 4.1 again, the left side equals

$$\big(2\theta\partial_\theta^2 + q\partial_\theta\big)L_\theta(T) = \big(2\theta a_T^2 - q a_T\big)L_\theta(T).$$

For the right side we need $\int W \operatorname{etr}(-TW)\,\mu_\theta(dW)$, $\theta \ge 0$. Write $W = xx^\intercal + V$ with $x = X_0 + \sqrt{\theta}v$. By (2.7), $\mathsf{E}[xx^\intercal e^{-x^\intercal T x}] = \mathsf{E}[e^{-x^\intercal T x}]\big(S_T + \theta B_T vv^\intercal B_T^\intercal\big)$. From the Wishart density, $\operatorname{etr}(-TY)\mathcal{W}_m(q-1,S)(dY) = |I+2ST|^{-(q-1)/2}\mathcal{W}_m(q-1,S_T)(dY)$, so $\mathsf{E}[V\operatorname{etr}(-TV)] = (q-1)S_T\,\mathsf{E}\operatorname{etr}(-TV)$. By independence,

$$\int W \operatorname{etr}(-TW)\,\mu_\theta(dW) = L_\theta(T)\big(qS_T + \theta B_T vv^\intercal B_T^\intercal\big).$$

By (2.2) with $H = vv^\intercal$, and since $v^\intercal B_T^\intercal T v = v^\intercal T B_T v = a_T$, the right side of (4.2) equals

$$L_\theta(T)\big(-q\,v^\intercal(T - 2TS_TT)v + 2\theta a_T^2\big) = L_\theta(T)\big(-qa_T + 2\theta a_T^2\big),$$

where we used (2.5). The two sides agree. □

Remark 4.3. If q is an integer, μ_θ is the law of $Y^\intercal Y$ for a $q \times m$ Gaussian matrix Y with independent $N_m(0,S)$ rows and mean $yv^\intercal$, where $y \in \mathbb{R}^q$ and $|y|^2 = \theta$, and $2\theta\partial_\theta^2 + q\partial_\theta$ is one half of the Laplacian in $y \in \mathbb{R}^q$, acting on functions of $\theta = |y|^2$.

Proposition 4.4. *Let $m = m_1 + \cdots + m_s$ be a partition with $\ell = \max_j m_j$, let $K_j \in \mathcal{S}_+^{m_j}$ and $b_j \ge 0$, and put $f(W) = \prod_j |W_{jj} + K_j|^{b_j}$ for $W \in \mathcal{S}_+^m$, where a factor with $b_j = 0$ is read as 1. If $q > m$ and $q \ge \ell + 1$, then $\theta \mapsto \int f\,d\mu_\theta$ is nondecreasing on $[0,\infty)$.*

Proof. Assume first that every K_j is positive definite, and let $\beta = \sum_j m_j b_j$. By Lemma 3.2 and Lemma 4.1, $g(\theta) = \int f\,d\mu_\theta$ is real analytic and g', g'' are given by integration against μ_θ', μ_θ''. Let $\tilde f$ be as in Lemma 3.2, let $\chi \in C_c^\infty(\mathcal{S}^m)$ satisfy $0 \le \chi \le 1$, $\chi(W) = 1$ for $\|W\| \le 1$ and $\chi(W) = 0$ for $\|W\| \ge 2$, and put $\varphi_R(W) = \chi(W/R)\tilde f(W)$ for $R \ge 1$. We apply (4.2) to φ_R and let $R \to \infty$. On the left, $|\varphi_R| \le |f|$ on $\mathcal{S}_+^m$, and dominated convergence gives $\int \varphi_R\,d\xi_\theta \to \int f\,d\xi_\theta = 2\theta g''(\theta) + qg'(\theta)$. On the right, since $\tilde f = f$ near $\mathcal{S}_+^m$, the product rule gives on $\mathcal{S}_+^m$

$$\mathcal{R}_{q,vv^\intercal}\varphi_R = \chi(W/R)\,\mathcal{R}_{q,vv^\intercal}f + E_R,$$

where E_R collects the terms containing at least one derivative of $\chi(W/R)$. These derivatives are $O(R^{-1})$ and $O(R^{-2})$ and vanish unless $R \le \|W\| \le 2R$, while the coefficients of $\mathcal{R}_{q,vv^\intercal}$ are of degree at most one in W. By Lemma 3.2, $|E_R(W)| \le c(1+\|W\|)^{\beta+1}\mathbf{1}_{\{\|W\|\ge R\}}$ and $|\mathcal{R}_{q,vv^\intercal}f| \le c(1+\|W\|)^{\beta+1}$ on $\mathcal{S}_+^m$. As μ_θ has moments of all orders, we conclude that

$$2\theta g''(\theta) + qg'(\theta) = \varrho(\theta) := \int \mathcal{R}_{q,vv^\intercal}f\,d\mu_\theta \qquad (\theta \ge 0). \tag{4.3}$$

By Lemma 3.1, $\varrho \geq 0$, and ϱ is continuous on $[0,\infty)$ by Lemma 4.1. For $\theta > 0$,

$$\frac{d}{d\theta}\big(\theta^{q/2}g'(\theta)\big) = \tfrac{1}{2}\theta^{q/2-1}\big(2\theta g''(\theta) + qg'(\theta)\big) = \tfrac{1}{2}\theta^{q/2-1}\varrho(\theta).$$

Since g' is continuous at 0 and $q > 0$, $\theta^{q/2}g'(\theta) \to 0$ as $\theta \downarrow 0$, and $t^{q/2-1}$ is integrable near 0. Hence

$$g'(\theta) = \tfrac{1}{2}\theta^{-q/2}\int_0^\theta t^{q/2-1}\varrho(t)\,dt \geq 0 \qquad (\theta > 0),$$

and g is nondecreasing on $[0,\infty)$.

In general, apply the first part to $f_\varepsilon(W) = \prod_j |W_{jj} + K_j + \varepsilon I|^{b_j}$, $0 < \varepsilon \leq 1$. For $W \in \mathcal{S}_+^m$ we have $f_\varepsilon(W) \to f(W)$ as $\varepsilon \downarrow 0$, and $0 \leq f_\varepsilon(W) \leq \prod_j (1 + \operatorname{tr} W + \operatorname{tr} K_j)^{m_j b_j}$. Dominated convergence at fixed θ passes the inequality $\int f_\varepsilon\,d\mu_{\theta_0} \leq \int f_\varepsilon\,d\mu_{\theta_1}$, $\theta_0 \leq \theta_1$, to the limit. Only the values of f_ε are passed to the limit, so no control of the derivatives of f near the boundary of the cone is needed; such derivatives may be unbounded when some b_j is small. □

5. Monotonicity for a fixed range

Proposition 5.1. *Let $m = m_1 + \cdots + m_s$ be a partition with $\ell = \max_j m_j$, let $S \in \mathcal{S}_{++}^m$, let $L \subseteq \mathbb{R}^m$ be a subspace of dimension r, and let α be real with*

$$q := \alpha - r + 1 > m \qquad \text{and} \qquad q \geq \ell + 1. \tag{5.1}$$

Let $\mathcal{P}_L = \{\Omega \in \mathcal{S}_+^m : \operatorname{ran}\Omega \subseteq L\}$, let $b_j \geq 0$, and put

$$h(\Omega) = \int \prod_{j=1}^s |W_{jj}|^{b_j}\,\mathcal{NW}_m(\alpha, S, \Omega)(dW), \qquad \Omega \in \mathcal{P}_L.$$

Then $h(\Omega_0) \leq h(\Omega_1)$ whenever $\Omega_0, \Omega_1 \in \mathcal{P}_L$ and $\Omega_0 \preceq \Omega_1$.

The law $\mathcal{NW}_m(\alpha, S, \Omega)$ exists for $\Omega \in \mathcal{P}_L$ by Lemma 2.3, since $\operatorname{rank}\Omega \leq r$ and $\alpha - r = q - 1 > m - 1$.

Proof. If $r = 0$, then $\mathcal{P}_L = \{0\}$. Let $r \geq 1$. We first show that $h(\Omega) \leq h(\Omega + c\,vv^{\mathsf{T}})$ for $\Omega \in \mathcal{P}_L$, $v \in L$ and $c \geq 0$. We may assume $v \neq 0$. We claim that

$$\Omega = \Omega' + \theta_0 vv^{\mathsf{T}} \qquad \text{with } \Omega' \in \mathcal{S}_+^m,\ \operatorname{ran}\Omega' \subseteq L,\ \operatorname{rank}\Omega' \leq r-1,\ \theta_0 \geq 0. \tag{5.2}$$

If $v \notin \operatorname{ran}\Omega$, then $\operatorname{ran}\Omega$ is a proper subspace of L; take $\Omega' = \Omega$ and $\theta_0 = 0$. If $v \in \operatorname{ran}\Omega$, write $v = \Omega^{1/2}w$ with $w \in \operatorname{ran}\Omega$, $w \neq 0$, so that $v^{\mathsf{T}}\Omega^{+}v = \|w\|^2$, and put $\theta_0 = \|w\|^{-2}$. Then $\Omega' = \Omega - \theta_0 vv^{\mathsf{T}} = \Omega^{1/2}P\Omega^{1/2}$, where P is the orthogonal projection onto $w^\perp$. Thus $\Omega' \succeq 0$, $\operatorname{ran}\Omega' \subseteq \operatorname{ran}\Omega$, and $\operatorname{rank}\Omega' = \operatorname{rank}(P\Omega^{1/2}) = \dim(\operatorname{ran}\Omega \cap w^\perp) = \operatorname{rank}\Omega - 1 \leq r - 1$.

Write $\Omega' = M'^{\mathsf{T}}M'$ with a real $(r-1)\times m$ matrix M', let Z' be an $(r-1)\times m$ matrix with independent $N_m(0,S)$ rows, and put $Y' = (Z'+M')^{\mathsf{T}}(Z'+M')$; for $r = 1$, $Y' = 0$. Let μ_θ be the laws of Section 4 with this S, v and q, and let $W_\theta \sim \mu_\theta$ be independent of Y'. By Lemma 2.3 and the additivity of types, $Y' + W_\theta$ is of type $(r-1+q, S, \Omega' + \theta vv^{\mathsf{T}}) = (\alpha, S, \Omega' + \theta vv^{\mathsf{T}})$, so by uniqueness

$$h(\Omega' + \theta vv^{\mathsf{T}}) = \mathsf{E}\big[g_{Y'}(\theta)\big], \qquad g_K(\theta) = \int \prod_j |W_{jj} + K_{jj}|^{b_j}\,\mu_\theta(dW).$$

For every fixed $K \in \mathcal{S}_+^m$, Proposition 4.4 applies with the shifts $K_{jj} \succeq 0$, because of (5.1); hence g_K is nondecreasing. Taking $\theta = \theta_0$ and $\theta = \theta_0 + c$ proves $h(\Omega) \le h(\Omega + c\,vv^{\mathsf{T}})$.

In general, $\Omega_1 - \Omega_0 \succeq 0$ has its range in L, so its spectral decomposition is $\sum_i c_i v_i v_i^{\mathsf{T}}$ with $c_i > 0$ and $v_i \in L$. All partial sums $\Omega_0 + \sum_{i \le t} c_i v_i v_i^{\mathsf{T}}$ belong to $\mathcal{P}_L$, and the rank-one case applies at each step. □

Remark 5.2. The decomposition (5.2) accounts for the value of q in (5.1). The part Ω', of rank at most $r-1$, is carried by $r-1$ Gaussian rows, so the rank-one part carries the shape $\alpha - r + 1$ rather than $\alpha - r$.

6. Separation of one block

Lemma 6.1. *Let $n = k + m$ with $k, m \ge 1$, let $\alpha > n - 1$, and let $\Sigma \in \mathcal{S}_{++}^n$ and $W \sim \mathcal{W}_n(\alpha, \Sigma)$ be partitioned as*

$$\Sigma = \begin{pmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{pmatrix}, \qquad W = \begin{pmatrix} W_{aa} & W_{ab} \\ W_{ba} & W_{bb} \end{pmatrix},$$

with diagonal blocks of sizes k and m. Put $K = \Sigma_{aa}^{-1}\Sigma_{ab}$ and $S = \Sigma_{bb} - \Sigma_{ba}\Sigma_{aa}^{-1}\Sigma_{ab}$. Then $W_{aa} \sim \mathcal{W}_k(\alpha, \Sigma_{aa})$, and for every Borel function $\varphi \ge 0$ on $\mathcal{S}_+^m$,

$$\mathsf{E}\big[\varphi(W_{bb}) \mid W_{aa}\big] = \int \varphi \, d\mathcal{NW}_m\big(\alpha, S, K^{\mathsf{T}} W_{aa} K\big) \qquad \text{almost surely,}$$

where the right side, which is defined for every $W_{aa} \in \mathcal{S}_+^k$, is a version of the conditional expectation.

Proof. Write $C = \Sigma_{aa}$, so that $\Sigma_{ab} = CK$ and $\Sigma_{bb} = S + K^{\mathsf{T}} C K$. Then

$$\Sigma = \begin{pmatrix} I & 0 \\ K^{\mathsf{T}} & I \end{pmatrix} \begin{pmatrix} C & 0 \\ 0 & S \end{pmatrix} \begin{pmatrix} I & K \\ 0 & I \end{pmatrix}, \qquad \Sigma^{-1} = \begin{pmatrix} C^{-1} + KS^{-1}K^{\mathsf{T}} & -KS^{-1} \\ -S^{-1}K^{\mathsf{T}} & S^{-1} \end{pmatrix}.$$

Write $W = \left(\begin{smallmatrix} A & E \\ E^{\mathsf{T}} & B \end{smallmatrix}\right)$ and change variables to (A, U, V) with $U = A^{-1/2}E$ and $V = B - E^{\mathsf{T}}A^{-1}E = B - U^{\mathsf{T}}U$. This is a bijection from $\mathcal{S}_{++}^n$ onto $\mathcal{S}_{++}^k \times \mathbb{R}^{k \times m} \times \mathcal{S}_{++}^m$; it is triangular, and for fixed A the map $U \mapsto A^{1/2}U$ multiplies each of the m columns by $A^{1/2}$, so the Jacobian is $|A|^{m/2}$. Moreover $|W| = |A|\,|V|$, and expanding $\operatorname{tr}(\Sigma^{-1}W)$ gives

$$\operatorname{tr}(\Sigma^{-1}W) = \operatorname{tr}(C^{-1}A) + \operatorname{tr}(S^{-1}V) + \operatorname{tr}\big(S^{-1}(U - A^{1/2}K)^{\mathsf{T}}(U - A^{1/2}K)\big).$$

Hence the density of (A, U, V) is proportional to

$$|A|^{(\alpha-k-1)/2} \operatorname{etr}\big(-C^{-1}A/2\big) \cdot |V|^{((\alpha-k)-m-1)/2} \operatorname{etr}\big(-S^{-1}V/2\big) \cdot \operatorname{etr}\big(-S^{-1}(U - A^{1/2}K)^{\mathsf{T}}(U - A^{1/2}K)/2\big).$$

Since $\alpha - k > m - 1$, the second factor is an unnormalized $\mathcal{W}_m(\alpha - k, S)$ density in V, and the third is an unnormalized Gaussian density in U whose normalizing constant does not depend on A. Integrating out U and V shows that A has a density proportional to the first factor, so $A \sim \mathcal{W}_k(\alpha, C)$; the conditional density of (U, V) given A is then the product of the last two factors, normalized. Hence, conditionally on A, the matrix $Z = U - A^{1/2}K$ has independent $N_m(0, S)$ rows and is independent of $V \sim \mathcal{W}_m(\alpha - k, S)$. Since

$$B = V + (Z + A^{1/2}K)^{\mathsf{T}}(Z + A^{1/2}K),$$

Lemma 2.3 shows that, conditionally on A, B is of type $(k+(\alpha-k), S, K^{\mathsf{T}}AK) = (\alpha, S, K^{\mathsf{T}}AK)$. □

Lemma 6.2. *Let $k \geq 1$, $\beta > k-1$ and $A \sim \mathcal{W}_k(\beta, I_k)$. Let $F, G : \mathcal{S}_+^k \to [0,\infty)$ be Borel functions that are nondecreasing in the Loewner order, and assume that $F(OXO^{\mathsf{T}}) = F(X)$ for all orthogonal O. If $\mathsf{E}[F(A)]$, $\mathsf{E}[G(A)]$ and $\mathsf{E}[F(A)G(A)]$ are finite, then $\mathsf{E}[F(A)G(A)] \geq \mathsf{E}[F(A)]\,\mathsf{E}[G(A)]$.*

Proof. Let $\lambda(X) = (\lambda_1 \leq \cdots \leq \lambda_k)$ denote the ordered eigenvalues of $X \in \mathcal{S}_+^k$, and let dO be the Haar probability measure on the orthogonal group $\mathrm{O}(k)$. For $\lambda \in [0,\infty)^k$ put

$$\tilde{F}(\lambda) = F(\mathrm{diag}(\lambda)), \qquad \bar{G}(\lambda) = \int_{\mathrm{O}(k)} G\big(O\,\mathrm{diag}(\lambda)O^{\mathsf{T}}\big)\,dO.$$

If $\lambda \leq \lambda'$ coordinatewise, then $O\,\mathrm{diag}(\lambda'-\lambda)O^{\mathsf{T}} \succeq 0$, so $\tilde{F}$ and $\bar{G}$ are nondecreasing in each coordinate. Writing $X = O_X\,\mathrm{diag}(\lambda(X))O_X^{\mathsf{T}}$ with $O_X \in \mathrm{O}(k)$, the invariance of F and of the Haar measure give $F(X) = \tilde{F}(\lambda(X))$ and $\int G(OXO^{\mathsf{T}})\,dO = \bar{G}(\lambda(X))$. The law of A is invariant under orthogonal conjugation. Hence, if O is Haar distributed and independent of A, the pair $(F(A), G(OAO^{\mathsf{T}})) = (F(OAO^{\mathsf{T}}), G(OAO^{\mathsf{T}}))$ has the law of $(F(A), G(A))$, and Fubini's theorem gives

$$\mathsf{E}[F(A)G(A)] = \mathsf{E}\big[\tilde{F}(\lambda)\bar{G}(\lambda)\big], \qquad \mathsf{E}[G(A)] = \mathsf{E}\big[\bar{G}(\lambda)\big], \qquad \lambda = \lambda(A). \tag{6.1}$$

The density of A is proportional to $\prod_i u(\lambda_i)$ with $u(t) = t^{(\beta-k-1)/2}e^{-t/2}$. By the Jacobian of the spectral decomposition [12, Theorem 3.2.17], λ has the density

$$\pi(\lambda) = c\prod_{i=1}^{k} u(\lambda_i)\prod_{i<j}(\lambda_j-\lambda_i)\,\mathbf{1}_{\mathcal{C}}(\lambda), \qquad \mathcal{C} = \{0 < \lambda_1 < \cdots < \lambda_k\}.$$

This density is multivariate totally positive of order two: $\pi(x\vee y)\pi(x\wedge y) \geq \pi(x)\pi(y)$ for $x, y \in \mathbb{R}^k$, where $\vee$ and $\wedge$ are the coordinatewise maximum and minimum. Indeed, if x or y lies outside $\mathcal{C}$ there is nothing to prove; $\mathcal{C}$ is closed under $\vee$ and $\wedge$; the factors $u(\lambda_i)$ cancel; and for each pair $i<j$ the gap factor satisfies the corresponding inequality. For the gap factor, the only nontrivial configuration is, up to exchanging x and y, $x_i = a \leq y_i = b < y_j = e \leq x_j = f$, in which case

$$(e-a)(f-b) - (f-a)(e-b) = (b-a)(f-e) \geq 0.$$

For related positive dependence properties of the eigenvalues of Wishart matrices, see [3] and the product inequality [7, Theorem 3.14]. Consequently λ is associated [4, 10]: $\mathsf{E}[\phi(\lambda)\psi(\lambda)] \geq \mathsf{E}\phi(\lambda)\,\mathsf{E}\psi(\lambda)$ for bounded Borel functions ϕ, ψ that are nondecreasing in each coordinate.

Applying association to $\min(\tilde{F}, t)$ and $\min(\bar{G}, t)$, extended to $\mathbb{R}^k$ by $\lambda \mapsto \min(\tilde{F}, t)(\lambda \vee 0)$ and similarly, and letting $t \to \infty$ by monotone convergence, we obtain $\mathsf{E}[\tilde{F}(\lambda)\bar{G}(\lambda)] \geq \mathsf{E}[\tilde{F}(\lambda)]\,\mathsf{E}[\bar{G}(\lambda)]$. By (6.1) this is the claim. □

Proposition 6.3. *Let $d \geq 2$, let $p_1, \ldots, p_d$ be positive integers with $p = p_1 + \cdots + p_d$, and let $W \sim \mathcal{W}_p(\alpha, \Sigma)$ be partitioned as in Theorem 1.1, where $\Sigma_{11} = I_{p_1}$. Let $\ell = \max_{i\geq 2} p_i$ and $r = \mathrm{rank}(\Sigma_{12}, \ldots, \Sigma_{1d})$, and assume*

$$\alpha > p-1 \qquad \text{and} \qquad \alpha \geq r + \ell. \tag{6.2}$$

Then (1.2) holds for all real $\nu_1, \ldots, \nu_d \geq 0$.

Proof. Apply Lemma 6.1 with $k = p_1$, $m = p - p_1$, $W_{aa} = W_{11}$ and W_{bb} the principal submatrix formed by the blocks $2, \dots, d$. Let K and S be as there; thus $K = (\Sigma_{12}, \dots, \Sigma_{1d})$, and $L = \operatorname{ran} K^{\mathsf{T}} \subseteq \mathbb{R}^m$ has dimension $r \le \min(k, m)$. Then $q = \alpha - r + 1 \ge \ell + 1$ by (6.2), and $q > (p-1) - r + 1 = k + m - r \ge m$. Hence (5.1) holds, with the partition of m into $p_2, \dots, p_d$ and $b_j = \nu_{j+1}$. By Lemma 6.1,

$$\mathsf{E}\Big[\prod_{i=2}^{d} |W_{ii}|^{\nu_i} \;\Big|\; W_{11}\Big] = G(W_{11}), \qquad G(X) = h(K^{\mathsf{T}} X K),$$

with h as in Proposition 5.1. For $X \in \mathcal{S}_+^k$ the range of $K^{\mathsf{T}} X K$ lies in L, and $X \preceq X'$ implies $K^{\mathsf{T}} X K \preceq K^{\mathsf{T}} X' K$. By Proposition 5.1, G is nondecreasing in the Loewner order; it is Borel by the representation in the proof of Lemma 6.1. Since $W_{11} \sim \mathcal{W}_k(\alpha, I_k)$ with $\alpha > k - 1$, and since the expectations involved are finite by Section 2, Lemma 6.2 applied to $F(X) = |X|^{\nu_1}$ and G gives

$$\mathsf{E}\Big[\prod_{i=1}^{d} |W_{ii}|^{\nu_i}\Big] = \mathsf{E}\big[F(W_{11})G(W_{11})\big] \ge \mathsf{E}\big[|W_{11}|^{\nu_1}\big]\,\mathsf{E}\big[G(W_{11})\big],$$

and $\mathsf{E}[G(W_{11})] = \mathsf{E}[\prod_{i \ge 2} |W_{ii}|^{\nu_i}]$. □

7. Proofs of the main results

Proof of Theorem 1.2. Let M be the block diagonal matrix with diagonal blocks $\Sigma_{11}^{-1/2}, \dots, \Sigma_{dd}^{-1/2}$. Then $MWM \sim \mathcal{W}_p(\alpha, M\Sigma M)$, the diagonal blocks of $M\Sigma M$ are identity matrices, and $|(MWM)_{ii}| = |\Sigma_{ii}|^{-1}|W_{ii}|$. Both sides of (1.1) and of (1.2) are multiplied by $\prod_i |\Sigma_{ii}|^{-\nu_i}$, so we may assume that $\Sigma_{ii} = I_{p_i}$ for all i; this does not change $\operatorname{rank} \Sigma_{ij}$. For $d = 1$ there is equality in (1.1). For $d \ge 2$, let ℓ and r be as in Proposition 6.3, so that $r \le p_1$.

(a) Let $\alpha \ge p$. We use induction on d; for $d = 1$ there is equality. Let $d \ge 2$. Since $r + \ell \le p_1 + \ell \le p \le \alpha$, Proposition 6.3 gives (1.2). The principal submatrix formed by the blocks $2, \dots, d$ has the law $\mathcal{W}_{p-p_1}(\alpha, \Sigma')$, where Σ' is the corresponding principal submatrix of Σ, and $\alpha \ge p > p - p_1$. By the induction hypothesis, $\mathsf{E}[\prod_{i \ge 2} |W_{ii}|^{\nu_i}] \ge \prod_{i \ge 2} \mathsf{E}[|W_{ii}|^{\nu_i}]$, and (1.1) follows.

(b) Let $d \ge 3$. Since at least two blocks have indices $i \ge 2$, we have $\ell \le p - p_1 - 1$, hence $r + \ell \le p_1 + \ell \le p - 1 < \alpha$, and Proposition 6.3 gives (1.2). The principal submatrix formed by the blocks $2, \dots, d$ has order $p - p_1 \le p - 1 < \alpha$, so part (a) applies to it, and (1.1) follows.

(c) Let $d = 2$; then (1.2) is (1.1). If $p_1 \ne p_2$, we exchange the two blocks if necessary, so that $p_1 > p_2$; then $\ell = p_2$, $r \le p_2$ and $r + \ell \le 2p_2 \le p - 1 < \alpha$. If $p_1 = p_2$ and $\operatorname{rank} \Sigma_{12} < p_1$, then $r \le p_1 - 1$, $\ell = p_1$ and $r + \ell \le p - 1 < \alpha$. In both cases Proposition 6.3 gives (1.1). □

Proof of Theorem 1.1. For $d = 1$ there is equality. By Theorem 1.2, it remains to consider $d = 2$, $p_1 = p_2$, Σ_{12} invertible and $p - 1 < \alpha < p$. In this case (1.1) is [9, Corollary 3.4] with $T = 0$ and $\nu_0 = 0$; the admissibility condition $0 > -\alpha/2 + (p-1)/2$ there is $\alpha > p - 1$. □

Proposition 6.3 applies after any simultaneous permutation of the blocks. This gives the following refinement.

Corollary 7.1. *In the setting of Theorem 1.1, assume that $d \geq 3$. Then for every $i \in \{1, \dots, d\}$,*

$$\mathsf{E}\Big[\prod_{j=1}^{d} |W_{jj}|^{\nu_j}\Big] \geq \mathsf{E}\big[|W_{ii}|^{\nu_i}\big]\, \mathsf{E}\Big[\prod_{j \neq i} |W_{jj}|^{\nu_j}\Big].$$

Remark 7.2. For $d = 2$, Theorem 1.2 proves (1.1) without the hypergeometric formula of [9], except when $p_1 = p_2$, Σ_{12} is invertible and $p - 1 < \alpha < p$. It gives the qualitative inequality (1.1) only, not the quantitative bound of [9, Corollary 3.4]. In the exceptional case $r = \ell = p_1$ whichever block is separated, and the condition $\alpha \geq r + \ell = p$ of Proposition 6.3 fails. For $d \geq 3$, the first separation leaves at least two blocks, and every later separation takes place in a principal submatrix of order at most $p - 1$.

Theorem 1.1 also removes the conditional hypothesis from a mixed-sign inequality of Genest, Ouimet and Richards; for $d = 2$ this was noted in [7, Remark 3.10].

Corollary 7.3. *In the setting of Theorem 1.1, put $P_{1i} = \Sigma_{11}^{-1/2} \Sigma_{1i} \Sigma_{ii}^{-1/2}$ for $i \geq 2$. If $(p_1 - 1)/2 < \nu_1 < \alpha/2 - (p_1 - 1)/2$ and $\nu_2, \dots, \nu_d > 0$, then*

$$\mathsf{E}\Big[|W_{11}|^{-\nu_1} \prod_{i=2}^{d} |W_{ii}|^{\nu_i}\Big] \geq \prod_{i=2}^{d} \big|I_{p_i} - P_{1i}^{\mathsf{T}} P_{1i}\big|^{\nu_i}\, \mathsf{E}\big[|W_{11}|^{-\nu_1}\big] \prod_{i=2}^{d} \mathsf{E}\big[|W_{ii}|^{\nu_i}\big].$$

Proof. This is [7, Proposition 3.9], which is proved there under the assumption that [7, Conjecture 1.1] holds. Its proof uses the conjecture only for a Wishart matrix of order p with α degrees of freedom and the exponent 0 on the first block, that is, inequality (1.1) for the principal submatrix formed by the blocks $2, \dots, d$. This submatrix is a Wishart matrix of order $p - p_1 < \alpha$, so Theorem 1.2(a) applies to it; for $d = 2$ there is nothing to prove. □

Remark 7.4. If $p_1 = \cdots = p_d = 1$ and $\alpha = n$ is an integer, (1.1) can be derived from [13, Theorem 1] as follows. Write $W = Y^{\mathsf{T}} Y$, where Y is an $n \times d$ matrix with independent $N_d(0, \Sigma)$ rows, and let Y_i be its columns, so that $W_{ii} = \|Y_i\|^2$. Let $U_1, \dots, U_d$ be independent and uniformly distributed on the unit sphere of $\mathbb{R}^n$, independent of Y. Then $\mathsf{E}_U |U_i^{\mathsf{T}} Y_i|^{2\nu_i} = c_i \|Y_i\|^{2\nu_i}$ with $c_i = \mathsf{E}|U_{i,1}|^{2\nu_i} > 0$. Conditionally on U, the vector $(U_i^{\mathsf{T}} Y_i)_i$ is centered Gaussian with variances Σ_{ii}, so [13, Theorem 1] gives $\mathsf{E}[\prod_i |U_i^{\mathsf{T}} Y_i|^{2\nu_i} \mid U] \geq \prod_i \mathsf{E}|N(0, \Sigma_{ii})|^{2\nu_i}$, a bound that does not depend on U. Since $U_i^{\mathsf{T}} Y_i \sim N(0, \Sigma_{ii})$, the same identity gives $\mathsf{E}|N(0, \Sigma_{ii})|^{2\nu_i} = c_i \mathsf{E}[|W_{ii}|^{\nu_i}]$, and dividing by $\prod_i c_i$ yields (1.1).

Remark 7.5. For nonnegative integer exponents, both sides of (1.1) are explicitly computable by the recursion of [6], whose degree-lowering operator is $\mathcal{R}_{\alpha,\Sigma}$. Lemma 4.2 may be viewed as a noncentral, rank-one analogue of the Wishart Stein identity, and Lemma 3.1 identifies a class of functions on which $\mathcal{R}_{q, vv^{\mathsf{T}}}$ is positive. For negative exponents and further mixtures of signs we refer to [7].

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

References

[1] G. Bailly, R. E. Gaunt, F. Ouimet, D. Richards, and R. von Sachs, *Stein's method for the Wishart distribution*, preprint (2026), arXiv:2606.04859.

[2] C. Benítez, Y. Sarantopoulos, and A. Tonge, *Lower bounds for norms of products of polynomials*, Math. Proc. Cambridge Philos. Soc. **124** (1998), no. 3, 395–408.

[3] R. L. Dykstra and J. E. Hewett, *Positive dependence of the roots of a Wishart matrix*, Ann. Statist. **6** (1978), no. 1, 235–238.

[4] C. M. Fortuin, P. W. Kasteleyn, and J. Ginibre, *Correlation inequalities on some partially ordered sets*, Comm. Math. Phys. **22** (1971), 89–103.

[5] P. E. Frenkel, *Pfaffians, hafnians and products of real linear functionals*, Math. Res. Lett. **15** (2008), no. 2, 351–358.

[6] R. E. Gaunt and F. Ouimet, *On Wilks' problem: exact recursive formulas via Stein's method for the joint moments of disjoint principal minors of Wishart random matrices*, preprint (2026), arXiv:2607.03984.

[7] C. Genest, F. Ouimet, and D. Richards, *On the Gaussian product inequality conjecture for disjoint principal minors of Wishart random matrices*, Electron. J. Probab. **29** (2024), Paper No. 166, 26 pp.

[8] C. Genest, F. Ouimet, and D. Richards, *On Wilks' joint moment formulas for embedded principal minors of Wishart random matrices*, Stat **13** (2024), no. 2, Paper No. e706, 6 pp.

[9] C. Genest, F. Ouimet, and D. Richards, *An explicit Wishart moment formula for the product of two disjoint principal minors*, Proc. Amer. Math. Soc. **153** (2025), no. 3, 1299–1311.

[10] S. Karlin and Y. Rinott, *Classes of orderings of measures and related correlation inequalities. I. Multivariate totally positive distributions*, J. Multivariate Anal. **10** (1980), no. 4, 467–498.

[11] W. V. Li and A. Wei, *A Gaussian inequality for expected absolute products*, J. Theoret. Probab. **25** (2012), no. 1, 92–99.

[12] R. J. Muirhead, *Aspects of Multivariate Statistical Theory*, John Wiley & Sons, New York, 1982.

[13] F. Ouimet and D. Greaves, *A proof of the strong Gaussian product inequality conjecture*, preprint, arXiv:2609.20234.

[14] S. S. Wilks, *Moment-generating operators for determinants of product moments in samples from a normal system*, Ann. of Math. (2) **35** (1934), no. 2, 312–340.

Knowledge Sense
Email address: `math@knowledgesense.jp`